

Supplement to: *Intermediate abundance promotes speciation when dispersal is limited*

S1 Computation set-up

We make use of a custom R package *abundolism* which, for review purposes, is hosted on anonymous github: <https://anonymous.4open.science/api/repo/abundolism-2DC5>

It can be installed from anonymous github, but note that dependencies on *Rcpp* and *RcppArmadillo* can lead to installation challenges on some infrastructure configurations.

```
tmp <- tempfile(fileext = ".zip")
download.file(
  "https://anonymous.4open.science/api/repo/abundolism-2DC5/zip",
  destfile = tmp
)

remotes::install_local(tmp)
```

The following R computing environment is used for all simulations and analyses:

```
library(abundolism)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
library(ggpointdensity)
library(cowplot)
library(knitr)
library(glmTMB)

sessionInfo() |>
  print(locale = FALSE)
```

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.0.1
```

Matrix products: default

```
BLAS:    /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/vecLib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; 1
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] glmmTMB_1.1.14      knitr_1.50          cowplot_1.2.0.9000
[4] ggpointdensity_0.2.1 ggplot2_4.0.1       dplyr_1.2.1
[7] abundolism_1.0.0
```

loaded via a namespace (and not attached):

[1]	sandwich_3.1-3	generics_0.1.4	lattice_0.22-7
[4]	lme4_1.1-38	digest_0.6.39	magrittr_2.0.4
[7]	evaluate_1.0.4	grid_4.5.2	RColorBrewer_1.1-3
[10]	fastmap_1.2.0	jsonlite_2.0.0	Matrix_1.7-4
[13]	mgcv_1.9-3	scales_1.4.0	numDeriv_2016.8-1.1
[16]	reformulas_0.4.4	Rdpack_2.6.4	cli_3.6.5
[19]	rlang_1.2.0	rbibutils_2.3	splines_4.5.2
[22]	withr_3.0.2	yaml_2.3.10	tools_4.5.2
[25]	nloptr_2.2.1	minqa_1.2.8	boot_1.3-32
[28]	vctrs_0.7.3	R6_2.6.1	zoo_1.8-14
[31]	lifecycle_1.0.5	MASS_7.3-65	pkgconfig_2.0.3
[34]	pillar_1.11.0	gtable_0.3.6	glue_1.8.0

[37]	Rcpp_1.1.0	xfun_0.52	tibble_3.3.0
[40]	tidyselect_1.2.1	rstudioapi_0.17.1	farver_2.1.2
[43]	htmltools_0.5.8.1	nlme_3.1-168	rmarkdown_2.29
[46]	TMB_1.9.21	compiler_4.5.2	S7_0.2.1

The quarto (Allaire et al. 2025) document used to generate this supplement can be found with the R command `system.file("ab_supp.qmd", package = "abundolism")`.

S2 Additional simulation details

The birth-death-immigration-speciation simulation tracks the following:

- all local population sizes
- running mean population size per iteration
- time relative to all rates (i.e. not simply number of iterations)
- whether an incipient speciation event has occurred
- the wait time to full speciation
- whether full speciation has occurred

In each iteration of the simulation an event is chosen at random. Possible events for each local population are

- local birth
- local death
- local speciation
- local dispersal

There is also one possible global event: dispersal from the source pool.

If there are n_p local populations then the total number of possible events are $n_p \times 4 + 1$. Which event occurs during any given simulation iteration is chosen at random according to the probability of each event. With a local population size of x_i , the probability of each event is proportional to its respective rate:

$$\mathbb{P}(\text{birth in pop. } i) = \frac{\lambda x_i}{R} \quad (1)$$

$$\mathbb{P}(\text{death in pop. } i) = \frac{\mu x_i}{R} \quad (2)$$

$$\mathbb{P}(\text{speciation in pop. } i) = \frac{\nu x_i}{R} \quad (3)$$

$$\mathbb{P}(\text{local emigration from pop. } i) = \frac{m_{prop} \gamma x_i}{R} \quad (4)$$

$$\mathbb{P}(\text{global emigration}) = \frac{\gamma}{R} \quad (5)$$

where $R = \gamma + (\sum_i^{n_p} x_i) (\lambda + \mu + \nu + m_{prop} \gamma)$ is a normalization constant.

In the case of dispersal (both local and from the source pool), the recipient of the dispersing individual must also be chosen. We choose the recipient local population at random with equal probability across local populations.

Once an event is randomly sampled, the variables being tracked are updated. Birth and immigration (local or global) augment a local population by 1. Death decrements a local population by 1. Dispersal events have no impact on the source of the dispersing individual. Time is updated following the Gillespie algorithm: the sojourn time between events is a random variable with an exponential probability distribution with rate R : $s \sim \text{Exp}(R)$. Each time step a random sojourn time is sampled and added to the running total time since the start of the simulation.

If a population experiences local extirpation all of its speciation-related information is re-set including removing any incipient speciation and restoring the wait time to full speciation back to the default of τ .

If full speciation is achieved the simulation stops. If no speciation occurs but the specified number of simulation iterations is reached, the simulation stops. Once stopped, the simulation returns the tracked variables of time, mean population size (i.e. abundance), and whether full speciation has occurred.

S3 Simulation experiment details

Simulation parameters are drawn from uniform distributions with the following minimum and maximum limits:

```
par_range <- data.frame(
  param = c("la", "mu", "g", "m_prop", "nu", "tau", "xi"),
  min    = c(0.1, 0.1, 0.1, 0, 0, 20, 10),
```

```

    max    = c(4,    4,    1,    0.1,    0.1, 200, 100)
)

# fixed params
np <- 10
nstep <- 10000

# number of simulation replicates
nrep <- 5000

# format param names for printing
mutate(par_range,
       param = sprintf("`%s`", param)) |>
  kable(format.args = list(scientific = FALSE))

```

param	min	max
la	0.1	4.0
mu	0.1	4.0
g	0.1	1.0
m_prop	0.0	0.1
nu	0.0	0.1
tau	20.0	200.0
xi	10.0	100.0

The parameters governing number of populations (**np**) and number of time steps (**nstep**) do not vary across simulations and are set to **np** = 10 and **nstep** = 10^4 . We simulate a total of **nrep** = 5000 replicates, each with a unique, randomly sampled set of parameter values.

S3.1 Running the simulation experiment

Parameter values for each simulation replicate are stored in a **data.frame** with one column for each parameter.

```

# change or remove for different results
set.seed(123)

# data.frame of randomly sampled param values
# one param per column

```

```

pars <- sapply(par_range$param, function(p) {
  runif(nrep,
        par_range[par_range$param == p, "min"],
        par_range[par_range$param == p, "max"])
}) |>
  as.data.frame()

```

With those parameters sampled, we can now run the simulation, one realization for each parameter combination. For details on inputs and outputs of the simulation function `sim_BDI_spec` see the help documentation accessible with `?sim_BDI_spec`.

```

# now run the simulation
sim_dat <- sim_BDI_spec(la = pars$la, mu = pars$mu, g = pars$g,
                      m_prop = pars$m_prop, nu = pars$nu,
                      tau = pars$tau, xi = pars$xi,
                      np = np, nstep = nstep, w = 100)

```

Simulation outputs are analyzed in binned windows of timesteps, in this case $w = 100$ steps in width. Bins are non-overlapping.

S4 Interpreting the simulation experiment

Now we can visually inspect the results. Our main purpose is to study how abundance relates to speciation in this model, but first we check other relationships to ensure the model behaves as expected. In particular, abundance should be larger when $\lambda > \mu$ and smaller when $\lambda < \mu$, supported primarily by immigration. When $\lambda > \mu$ populations will grow indefinitely. When birth and death rates are equal ($\lambda = \mu$) the population could crash or expand depending on small random fluctuations and random immigration. The figure below shows exactly this pattern. Smaller abundances for $\lambda > \mu$ likely correspond to time bin windows earlier in the simulation as well as those that stochastically drifted to smaller sizes.

```

fig_la_mu <- mutate(sim_dat,
                    disp_bin = cut(g * (1 + m_prop),
                                   breaks = c(-Inf, 0.33, 0.66, Inf))) |>
  ggplot(aes(la - mu, mean_pop_size)) +
  geom_hex() +
  facet_grid(
    cols = vars(speciation),
    labeller = as_labeller(c(`0` = "No speciation",
                             `1` = "Yes speciation"))
  )

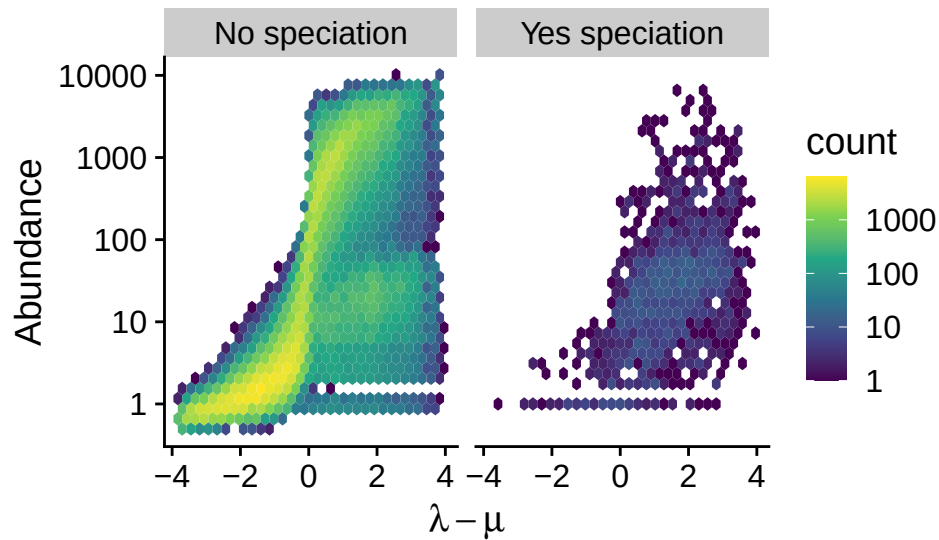
```

```

) +
  scale_y_log10() +
  xlab(expression(italic(lambda - mu))) +
  ylab("Abundance") +
  scale_fill_viridis_c(transform = "log10") +
  theme_cowplot()

fig_la_mu

```



Supplementary Figure S1: Relationship across simulation run windows between growth rate ($\lambda - \mu$) and abundance. Panels show simulation runs without and with full speciation. Colored hexagons show the number of points with values in that small neighborhood in order to more clearly show the distribution of simulation results.

```

fig_g <- mutate(sim_dat,
  growth_bin = cut(la - mu,
    breaks = c(-Inf, -2, 0, 2, Inf),
    labels = c("lo", "mid_lo", "mid_hi", "hi"))) |>
  mutate(growth_bin = factor(growth_bin,
    levels = c("hi", "mid_hi", "mid_lo", "lo"))) |>
  ggplot(aes(g * (1 + m_prop), mean_pop_size)) +
  geom_hex(bins = 30) +
  facet_grid(rows = vars(growth_bin),
    cols = vars(speciation),
    scales = "free_y",

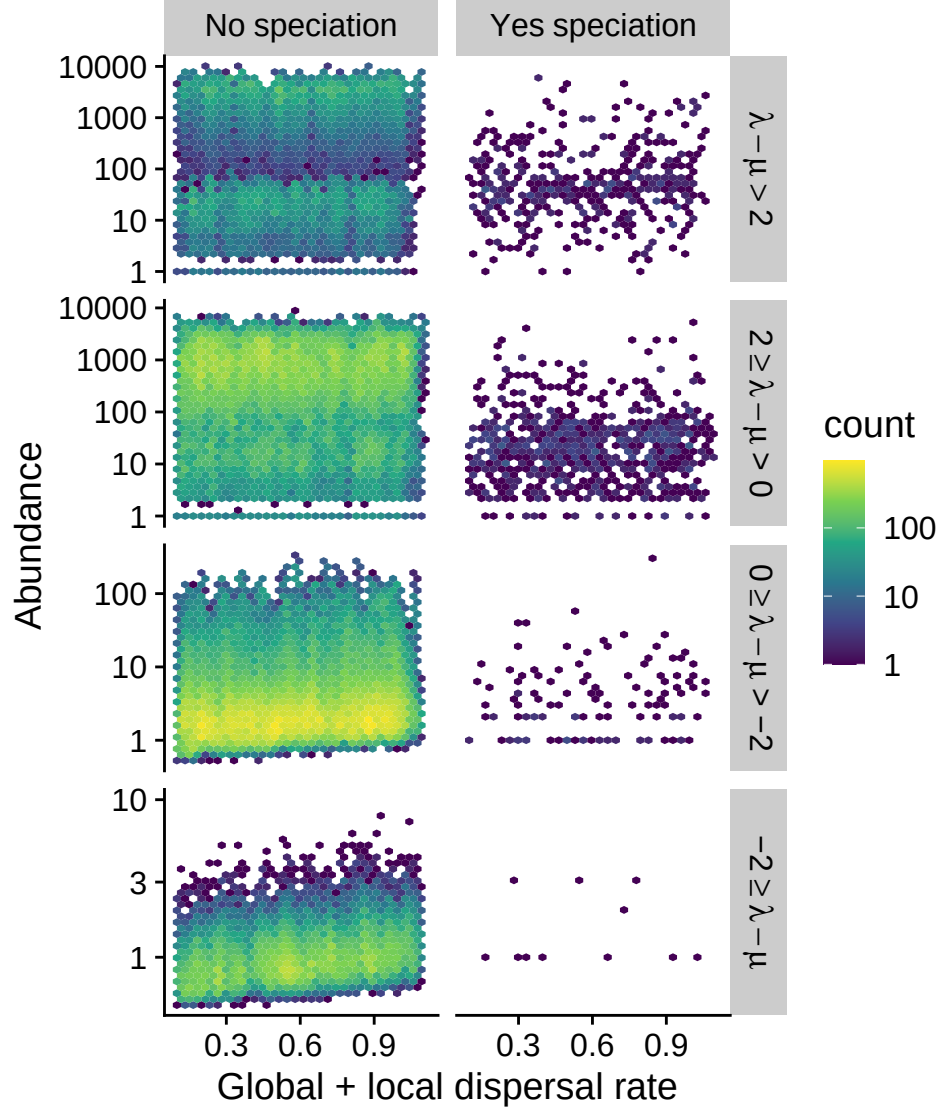
```

```

labeller = as_labeller(c(`0` = "No~speciation",
                          `1` = "Yes~speciation",
                          lo = "\"-2\">=lambda - mu",
                          mid_lo = "0 >= lambda - mu~\"> -2\"",
                          mid_hi = "2 >= lambda - mu~\"> 0\"",
                          hi = "lambda - mu > 2"),
                      label_parsed)) +
scale_y_log10(limits = function(y) {
  if(max(y) < 10) {
    return(c(min(y), 10))
  } else {
    return(range(y))
  }
}) +
xlab(expression("Global + local dispersal rate")) +
ylab("Abundance") +
scale_fill_viridis_c(transform = "log10") +
theme_cowplot()

```

fig_g



Supplementary Figure S2: Relationship across simulation run windows between combined dispersal rate (global plus local) and abundance. Panel columns show simulation run windows without and with full speciation while panel rows show simulation run windows of different growth rate ($\lambda - \mu$). Colored hexagons show the number of points with values in that small neighborhood in order to more clearly show the distribution of simulation results.

S4.1 Evaluating the relationship between abundance and speciation

These results are reported in the main text. Here we show how the results from the main text are derived.

First we build a quadratic model of probability of speciation as a function of abundance with the below code:

```
sim_mod <- glmmTMB(speciation ~ log_abund + I(log_abund^2) + (1 | sim_id),  
                  family = binomial, data = sim_dat)  
  
sim_quad_ci <- confint(sim_mod, parm = 3)
```

The confidence interval for the quadratic term is far from 0:

```
sim_quad_ci
```

	2.5 %	97.5 %	Estimate
I(log_abund^2)	-0.3109405	-0.2573601	-0.2841503

It should also be noted that the confidence interval for the quadratic term for abundance does not overlap zero in any of the GLMMs that included simulation model parameters as covariates along with abundance (see figure below).

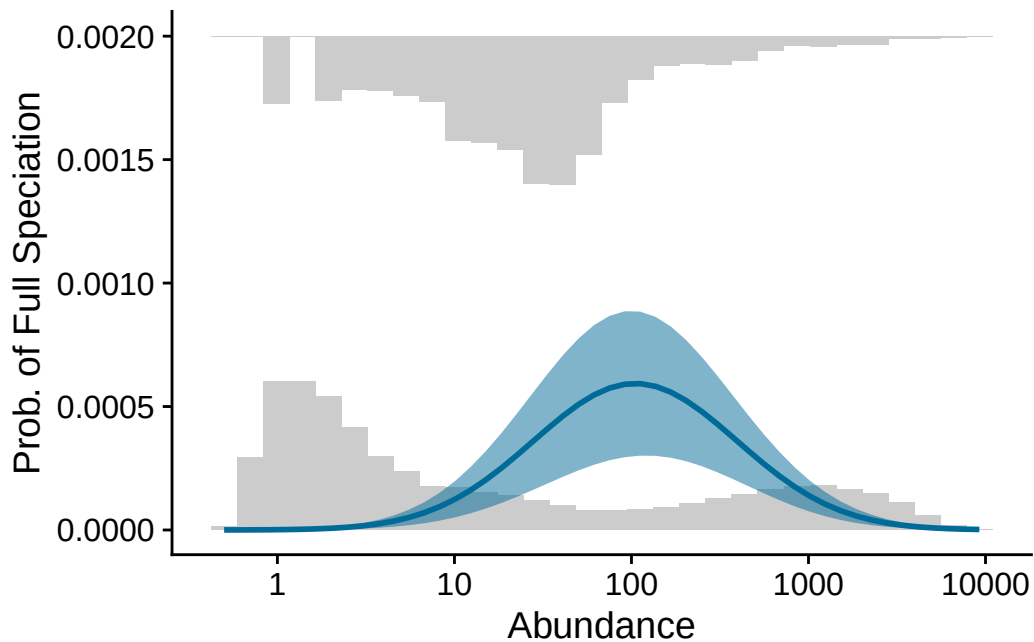
We now turn to developing the figure of abundance versus speciation. These results are reported in the main text, but the code below is used to create the visualization of speciation versus abundance.

```
# model prediction  
sim_pred <- pred_binom(sim_mod) |>  
  mutate(abund = exp(log_abund))  
  
# base plot of simulation results  
sim_fig_base <- logistic_hist_plot(sim_dat, mean_pop_size, xlog = TRUE,  
                                  xlab = "Abundance",  
                                  ymin = 0, ymax = 0.002)  
  
# add glmm CI and prediction  
sim_fig <- sim_fig_base +
```

```

geom_ribbon(data = sim_pred,
           mapping = aes(abund,
                         ymin = fit - 2 * se_fit,
                         ymax = fit + 2 * se_fit),
           fill = "#006B99", alpha = 0.5,
           inherit.aes = FALSE) +
geom_line(data = sim_pred,
          mapping = aes(abund, fit),
          color = "#006B99",
          linewidth = 1,
          inherit.aes = FALSE)
sim_fig

```



Supplementary Figure S3: Relationship between abundance and probability of full speciation. This figure is presented and discussed in more detail in the main text.

The fact that the peak of the GLMM does not clearly align with the peak of the speciation histogram is both because the peak captures the proportion of speciating to non-speciating windows and because the model accounts for random effects.

The mean abundance, and its confidence interval, at which speciation probability peaks can be calculated as

```

# MLE of peak location
abund_coef <- fixef(sim_mod)[[1]][2:3]
log_abund_peak <- -abund_coef[1] / (2 * abund_coef[2])

# var cov matrix of abund_coef
V <- vcov(sim_mod)$cond[2:3, 2:3]

# gradient
grad <- c(-1 / (2 * abund_coef[2]), abund_coef[1] / (2 * abund_coef[2]^2))

# ci calc
log_peak_se <- as.numeric(t(grad) %*% V %*% grad) |>
  sqrt()

log_peak_ci <- log_abund_peak + c(-1, 1) * 1.96 * log_peak_se

# putting things on a linear scale
exp(log_abund_peak)

```

```

log_abund
105.0223

```

```

exp(log_peak_ci)

```

```

[1] 88.85298 124.13402

```

S4.2 Assessing the interaction of abundance with simulation model parameters

We now evaluate the effects of simulation model parameters interacting with emergent simulated abundance to predict the probability of speciation. We look at how the linear and quadratic terms for mean abundance interact with ν , the per capita rate of incipient speciation; τ , the base wait time to full speciation; the combined dispersal rate (per capita local + global); and the per capita growth rate ($\lambda - \mu$).

We evaluate the relationships between each of these model parameters and the presence or absence of speciation in simulation run windows by fitting binomial generalized linear mixed effects models (GLMMs, fit using the *glmmTMB* package McGillicuddy et al. 2025).

```

# add columns with computed variables
sim_dat <- mutate(sim_dat,
  log_abund = log(mean_pop_size),
  tot_disp = g * (1 + m_prop),
  growth = la - mu)

# nu model
nu_mod <- glmmTMB(speciation ~
  (log_abund + I(log_abund^2)) *
  scale(nu) +
  (1 | sim_id),
  family = binomial, data = sim_dat)

nu_ci <- confint(nu_mod)

# tau model
tau_mod <- glmmTMB(speciation ~
  (log_abund + I(log_abund^2)) *
  scale(tau) +
  (1 | sim_id),
  family = binomial, data = sim_dat)

tau_ci <- confint(tau_mod)

# relative dispersal model
disp_mod <- glmmTMB(speciation ~
  (log_abund + I(log_abund^2)) *
  scale(tot_disp) +
  (1 | sim_id),
  family = binomial, data = sim_dat)

disp_ci <- confint(disp_mod)

# growth rate model
growth_mod <- glmmTMB(speciation ~
  (log_abund + I(log_abund^2)) *
  scale(growth) +
  (1 | sim_id),
  family = binomial, data = sim_dat)

```

```
growth_ci <- confint(growth_mod)
```

The figure below shows estimated effect sizes and 95% CIs of the model parameters, along with the estimated quadratic term for abundance, whose 95% CI never overlaps 0 in any of these models.

```
# assumes simulation model covariates have been scaled and centered
delta_peak <- function(b1, b2, b1_int, b2_int) {
  -(b1 + b1_int) / (2 * (b2 + b2_int))
}

cis <- list(
  nu_ci,
  tau_ci,
  disp_ci,
  growth_ci
) |>
  lapply(function(x) {

    # quadratic abundance and covariate coeffs
    params <- x[3:4, ]

    # shift in peak
    d_peak <- delta_peak(x[2, 3], x[3, 3], x[5, 3], x[6, 3])

    out <- rbind(params,
      # first 2 elements of below are not CI but used for
      # plot limits
      cbind(exp(d_peak) - exp(log_abund_peak),
        0,
        d_peak))

    # add columns
    out <- as.data.frame(out) |>
      mutate(param = rownames(params)[2],
        type = c("quad_abund", "covar", "delta_peak"))

    rownames(out) <- NULL
    out
  }) |>
```

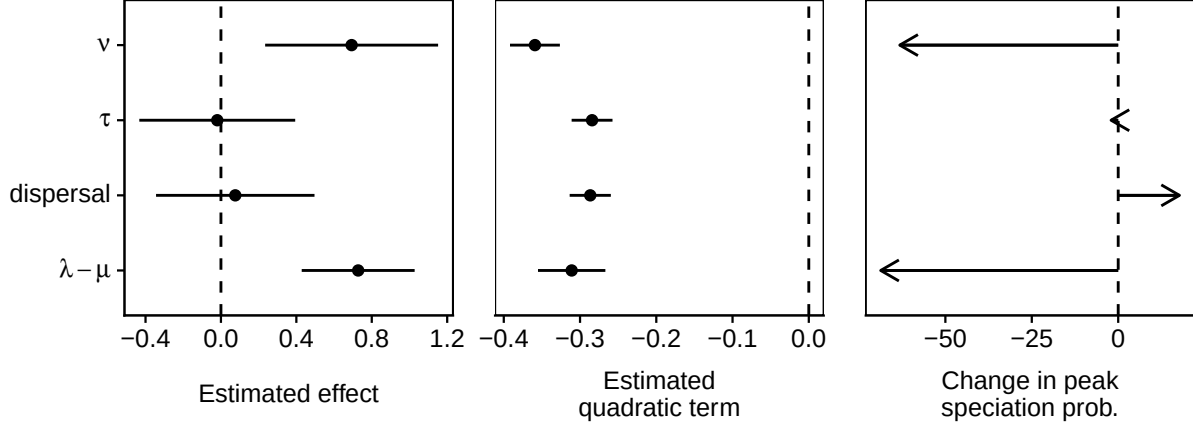
```

bind_rows() |>
mutate(param = factor(gsub("scale\\(|\\)", "", param),
                      levels = c("growth", "tot_disp", "tau", "nu")),
       type = factor(type,
                     levels = c("covar", "quad_abund", "delta_peak"))) |>
rename(ci_low = `2.5 %`,
       ci_hi = `97.5 %`)

fig_covar <- ggplot(cis, aes(x = ci_hi, xend = ci_low, param)) +
  geom_segment() +
  scale_y_discrete(labels = c(expression(lambda - mu),
                                expression("dispersal"),
                                expression(tau),
                                expression(nu))) +
  facet_grid(cols = vars(type), scales = "free_x",
             labeller = as_labeller(c(covar = "Estimated effect",
                                       quad_abund = "Estimated\nquadratic term",
                                       delta_peak = "Change in peak\nspeciation prob.")),
             switch = "x") +
  geom_vline(xintercept = 0, linetype = 2) +
  geom_segment(data = filter(cis, type == "delta_peak"),
              arrow = arrow(length = unit(10, "pt"))) +
  geom_point(data = filter(cis, type != "delta_peak"),
            mapping = aes(x = Estimate)) +
  theme_cowplot() +
  theme(panel.border = element_rect(color = "black", linewidth = 0.5),
        strip.placement = "outside",
        strip.background = element_blank(),
        axis.title = element_blank(),
        panel.spacing.x = unit(20, "pt"))

fig_covar

```



Supplementary Figure S4: Summary of different GLMMs predicting speciation probability. Different models are shown across the y-axis, labeled by the simulation model parameter included as a covariate along with the hump-shaped relationship effect of abundance. Estimated effect sizes and 95% confidence intervals are shown in the left hand panel. The quadratic term for abundance from each model is shown in the middle panel. The right hand panel shows the shift in abundance value at which probability of speciation reaches its peak. This shift is shown relative to the model without simulation model parameters as covariates. The magnitudes each arrow represent how the location of peak speciation probability would shift with an increase in the simulation model covariate.

S4.3 Assessing the effect of binning window size

In the main analysis we use a bin window size of 100 time steps. Here we re-run the analysis with windows of 50 or 200 to evaluate the impact of window size. Supplementary Figure S5 shows the intermediate abundance hump remains in the two alternative bin window sizes. As with the main analysis, the fact that the peak of the GLMM does not clearly align with the peak of the speciation histogram is both because the peak captures the proportion of speciating to non-speciating windows and because the model accounts for random effects.

```
# run the simulation with different window sizes
sim_dat_50 <- sim_BDI_spec(la = pars$la, mu = pars$mu, g = pars$g,
                          m_prop = pars$m_prop, nu = pars$nu,
                          tau = pars$tau, xi = pars$xi,
                          np = np, nstep = nstep, w = 50)

sim_dat_200 <- sim_BDI_spec(la = pars$la, mu = pars$mu, g = pars$g,
```

```

      m_prop = pars$m_prop, nu = pars$nu,
      tau = pars$tau, xi = pars$xi,
      np = np, nstep = nstep, w = 200)

# wrapper function to combine plotting steps
mk_window_plot <- function(dat, ymax = 0.002) {
  # add log abund
  dat$log_abund <- log(dat$mean_pop_size)

  # make model
  mod <- glmmTMB(speciation ~ log_abund + I(log_abund^2) + (1 | sim_id),
                 family = binomial, data = dat)

  # get pred and SE for plotting
  pred <- pred_binom(mod) |>
    mutate(abund = exp(log_abund))

  # base plot
  gp_base <- logistic_hist_plot(dat, mean_pop_size, xlog = TRUE,
                                xlab = "Abundance",
                                ymin = 0, ymax = ymax)

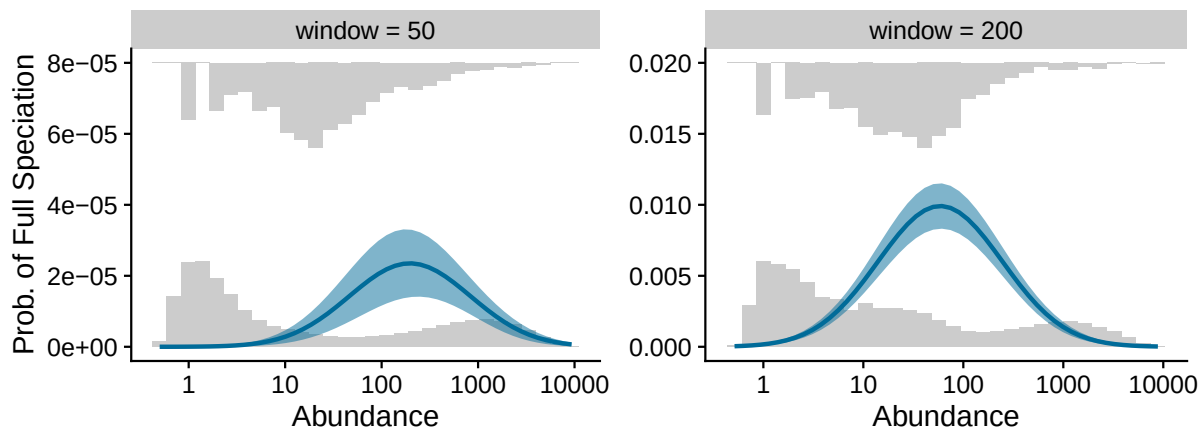
  gp_base +
    geom_ribbon(data = pred,
              mapping = aes(abund,
                            ymin = fit - 2 * se_fit,
                            ymax = fit + 2 * se_fit),
              fill = "#006B99", alpha = 0.5,
              inherit.aes = FALSE) +
    geom_line(data = pred,
             mapping = aes(abund, fit),
             color = "#006B99",
             linewidth = 1,
             inherit.aes = FALSE)
}

# make plots for each window size
w50 <- mk_window_plot(sim_dat_50, ymax = 0.00008) +
  facet_grid(cols = vars(w),
             labeller = as_labeller(c(`50` = "window = 50"))))

```

```
w200 <- mk_window_plot(sim_dat_200, ymax = 0.02) +
  ylab("") +
  theme(axis.title.y = element_blank()) +
  facet_grid(cols = vars(w),
             labeller = as_labeller(c(`200` = "window = 200"))))

# combine individual plots
plot_grid(w50, w200, rel_widths = c(1.05, 1))
```



Supplementary Figure S5: Effect of window size on relationship between abundance and full speciation within a window. In both cases of window size = 50 or 200, a mid-abundance hump in speciation probability is observed.

One thing to note in Supplementary Figure S5 is the difference in y-axis limits. This difference in limits reflects the fact that in smaller windows, the probability of encountering a speciation event is smaller, so the probability of full speciation (the y-axis) is overall lower in the `window = 50` panel compared to the `window = 200` panel.

S5 Analyzing abundance and richness of arthropod genera in Hawai‘i

The results of analyzing Hawai‘i arthropod data are presented in the main text. The code to produce these results is shown here.

S5.1 Preparing the data

First we load and prepare the data. Preparation involves filtering data to only endemic taxa and compiling average per-species abundance for genera based on data from Gruner (2007) and compiling species richness for genera based on the Bernice Pauahi Bishop Museum arthropod checklist (Nishida 2002).

```
# read gruner data
arth <- read.csv("raw_data/arth.csv")

# read checklist data
arth_check <- read.csv("raw_data/hawaii_arthropod_checklist.csv")

hmean <- function(x) length(x) / sum(1 / x)

# calculate average per-species abundances of genera from survey data
gen_abund <- filter(arth, origin == "native-endemic") |>
  group_by(genus, species_code) |>
  summarize(abund = n()) |>
  ungroup(species_code) |>
  summarize(mean_abund = mean(abund),
             survey_nspp = n_distinct(species_code)) |>
  ungroup()
```

`summarise()` has regrouped the output.
i Summaries were computed grouped by genus and species_code.
i Output is grouped by genus.
i Use `summarise(.groups = "drop\_last")` to silence this message.
i Use `summarise(.by = c(genus, species\_code))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.

```
# calculate richness of genera from checklist
gen_rich <- filter(arth_check, Genus != "") |>
  group_by(Genus) |>
  summarize(nspp = n_distinct(Genus.and.species),
             status = first(Status)) |>
  ungroup() |>
  rename(genus = Genus) |>
  # make `status` column indicate if genus is found in survey data
  mutate(status = ifelse(genus %in% gen_abund$genus,
                         "in survey",
                         status))
```

```
# `gen` holds data ready to analyze for richness ~ abundance
gen <- inner_join(gen_rich, gen_abund)
```

Joining with `by = join\_by(genus)`

S5.2 Model of within genus richness and average per-species abundance

With the prepared data we can compute a quadratic and linear model of genus richness versus average per-species abundance. As explained in the main text, these models are beta-geometric generalized linear models and are achieved with custom functions available in the *abundolism* package associated with this paper. The main function is `glm_bgeom` which maximizes the log likelihood function and shares some common methods with `lm` and `glm` including `logLik` which enables likelihood analyses such as the likelihood ratio test.

To summarize from the main text, our full quadratic model is

$$\begin{aligned} y_i &\sim \text{BetaGeometric}(\alpha_i, \beta) \\ \log(\alpha_i) &= a_0 + a_1 x_i + a_2 x_i^2 \\ \log(\beta) &= b_0 \end{aligned}$$

where x_i is average per-species abundance of lineage i and our estimate for the observable diversification probability for lineage i is $1 - \hat{p}_i = 1 - \frac{\alpha_i}{\alpha_i + \beta}$.

```
# add column for log abundance
gen$log_abund <- log(gen$mean_abund)

# quadratic model
arth_mod <- glm_bgeom(nspp ~ log_abund + I(log_abund^2), data = gen)

# straight line model
arth_mod0 <- glm_bgeom(nspp ~ log_abund, data = gen)

# view the model output
arth_mod
```

Call: `glm_bgeom(formula = nspp ~ log_abund + I(log_abund^2), data = gen)`

Coefficients:

(Intercept)	log_abund	I(log_abund^2)	(Intercept):beta
0.9204501	-0.5453639	0.1000263	2.9611979

```
Log likelihood:
'log Lik.' -493.7613 (df=4)
```

```
arth_mod0
```

```
Call: glm_bgeom(formula = nspp ~ log_abund, data = gen)
```

```
Coefficients:
      (Intercept)      log_abund (Intercept):beta
      0.4126521      -0.0545682       2.7655422
```

```
Log likelihood:
'log Lik.' -497.2671 (df=3)
```

We can also look at the confidence intervals for parameters of the quadratic model (note: these are Wald confidence intervals).

```
confint(arth_mod)
```

	2.5 %	97.5 %
(Intercept)	0.20710475	1.6337954
log_abund	-0.92232423	-0.1684037
I(log_abund^2)	0.03009513	0.1699574
(Intercept):beta	2.12655203	3.7958437

The likelihood ratio test comparing quadratic and linear models is achieved with a custom function `lrt_bgeom` from the *abundolism* package.

```
arth_lrt <- lrt_bgeom(arth_mod0, arth_mod)
```

As with the simulation results, we can calculate the average per-species abundance where predicted genus richness (and therefore $1 - \hat{p}$) is at its maximum

```
# MLE of peak location
arth_coef <- coef(arth_mod)[2:3]
log_arth_peak <- -arth_coef[1] / (2 * arth_coef[2])

# var cov matrix of abund coef
V_arth <- solve(-arth_mod$hess[2:3, 2:3])
```

```
# gradient
grad_arth <- c(-1 / (2 * arth_coef[2]), arth_coef[1] / (2 * arth_coef[2]^2))

# ci calc
log_arth_peak_se <- as.numeric(t(grad_arth) %*% V_arth %*% grad_arth) |>
  sqrt()

log_arth_peak_ci <- log_arth_peak + c(-1, 1) * 1.96 * log_arth_peak_se

# putting things on a linear scale
exp(log_arth_peak)
```

```
log_abund
15.27326
```

```
exp(log_arth_peak_ci)
```

```
[1] 8.64135 26.99493
```

To produce visualizations of the arthropod data we make use of a custom `ci_bgeom` function from *abundolism* that calculates confidence intervals and predictions from the output of `glm_bgeom`. This function first calculates standard errors for the $\log(\alpha)$ and $\log(\beta)$ terms of the beta-geometric GLM using the Wald approximation and then computes the standard error for $1 - \hat{p}$ using a change of variable approach. The predicted value $\hat{y}$ for numbers of species per genus is then $\hat{y} = \frac{1}{1-\hat{p}}$ and its standard error is calculated from that of $1 - \hat{p}$ again with a change of variable approach.

```
arth_mod_ci <- cbind(mean_abund = gen$mean_abund,
                     ci_bgeom(arth_mod, type = "response"))

arth_p_ci <- cbind(mean_abund = gen$mean_abund,
                   ci_bgeom(arth_mod, type = "prob"))

gp_nspp <- ggplot(arth_mod_ci, aes(mean_abund, ymin = y_lwr, ymax = y_upr)) +
  geom_ribbon(fill = "gray60", alpha = 0.5) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Average per-species abundance") +
  ylab("Genus richness") +
  geom_pointdensity(data = gen, mapping = aes(mean_abund, nspp),
```

```

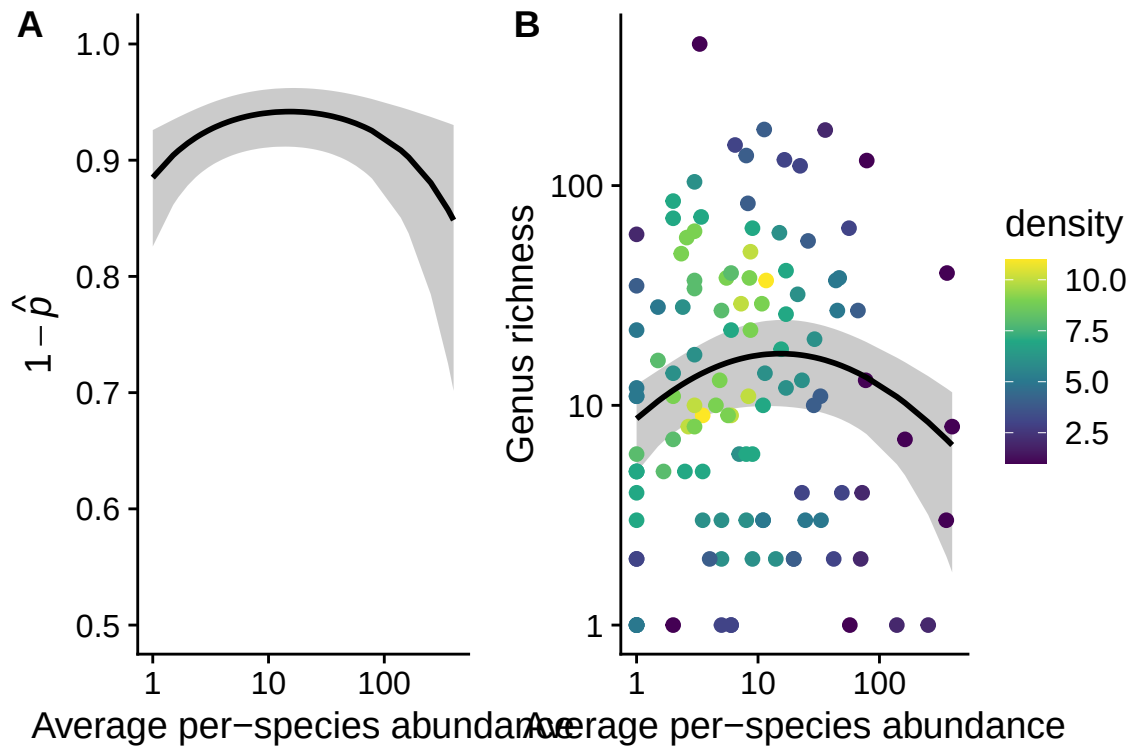
        inherit.aes = FALSE) +
scale_color_viridis_c() +
geom_line(data = arth_mod_ci,
          mapping = aes(mean_abund, y_hat),
          linewidth = 1) +
theme_cowplot()

gp_prb <- ggplot(arth_p_ci, aes(mean_abund, y = p_hat,
                               ymin = p_lwr, ymax = p_upr)) +
  geom_ribbon(fill = "gray60", alpha = 0.5) +
  geom_line(linewidth = 1) +
  scale_x_log10() +
  ylim(0.5, 1) +
  xlab("Average per-species abundance") +
  ylab(expression(italic(1 - hat(p)))) +
  theme_cowplot()

gp_arth <- plot_grid(gp_prb, gp_nspp, rel_widths = c(0.75, 1),
                    labels = c("A", "B"), label_x = 0.02)

gp_arth

```



S5.3 Evaluating impact of outliers

There is one clear outlier: *Drosophila* with high richness given abundance. To make sure this outlier does not bias results we re-run the analysis of genus richness versus average per-species abundance without *Drosophila*.

```
arth_nodro <- glm_bgeom(nspp ~ log_abund + I(log_abund^2),
  data = gen[gen$nspp < 400, ])
arth_nodro0 <- glm_bgeom(nspp ~ log_abund,
  data = gen[gen$nspp < 400, ])
```

Based on a likelihood ratio test we find the quadratic model is still preferred.

```
lrt_bgeom(arth_nodro0, arth_nodro)
```

Likelihood ratio test

Model 0: nspp ~ log_abund

Model 1: nspp ~ log_abund + I(log_abund^2)

```

      #Df  LogLik Df  Chisq Pr(>Chisq)
Model 0    3 -486.59
Model 1    4 -483.10  1 6.9828    0.00823 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

S5.4 Evaluating impact of missing genera

In merging the checklist data (providing species richness) with field survey data (providing abundance estimates) some genera were lost due to not being detected in the survey data.

```

# endemic checklist genera not found in dan s data
# only need to search for "end" because those with
# status "end" that are in gruner data had their
# status switched to "in survey"
n_checklist_missing <- sum(gen_rich$status == "end")
n_checklist_missing

```

```
[1] 479
```

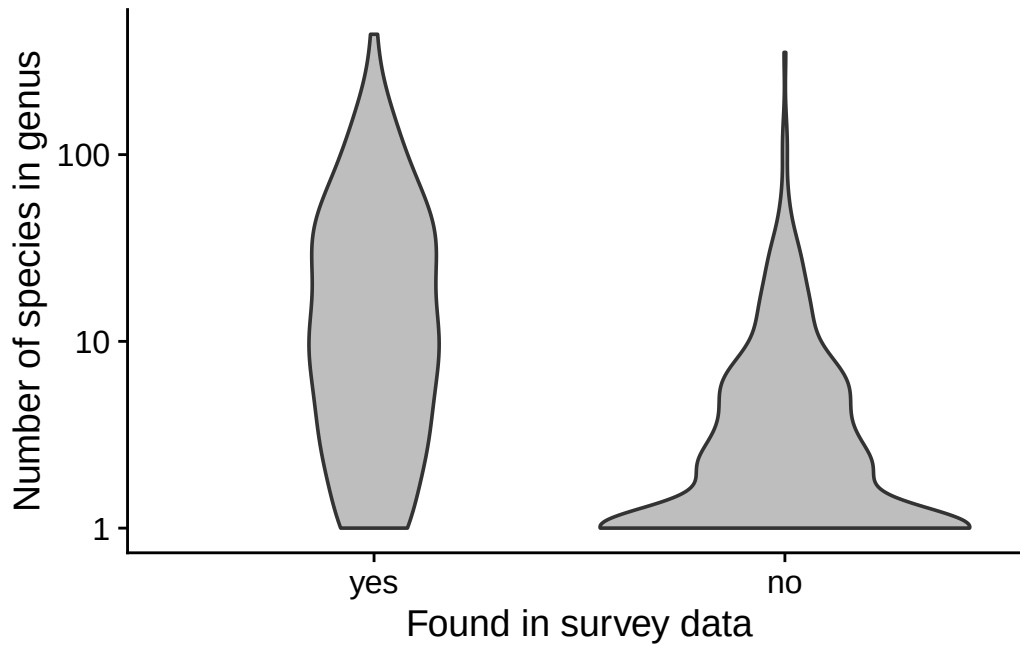
479 endemic genera from the checklist were absent from the survey data, representing 80.23% of endemic genera from the checklist.

While this is a large percentage, it is also not unexpected. The survey data constitute a very small sample of the geographic space of the Hawaiian Archipelago as well as a small sample of the extremely heterogeneous environmental space of the archipelago. It is still worth evaluating how the distributions of species richnesses differ between those genera included in the survey data and those missing. Supplementary Figure S6 shows that genera not found in the survey data are highly skewed toward those with low species richness.

```

filter(gen_rich, status %in% c("end", "in survey")) |>
  mutate(status = factor(ifelse(status == "end", "no", "yes"),
                           levels = c("yes", "no"))) |>
  ggplot(aes(status, nspp)) +
  geom_violin(fill = "gray") +
  scale_y_log10() +
  theme_cowplot() +
  xlab("Found in survey data") +
  ylab("Number of species in genus")

```



Supplementary Figure S6: Distribution of per genus richnesses for those genera found in the survey data (Gruner 2007) and those not found in the survey data.

Any evaluation of how these missing genera might change the relationship of average per-species abundance and genus richness (estimated in Section S5.2) will be compromised because it is impossible to know the abundances of the species in the missing genera. These genera could be absent from the survey data because their constituent species are rare (i.e. low average per-species abundance) or because their constituent species are restricted to different habitats and/or islands (i.e. potentially high average per-species abundance found elsewhere). However, for the sake of producing any kind of evaluation, we assume that if any of these missing genera were to be present in the survey data, they would be very rare. Thus, we re-run our analysis of the relationship between average per-species abundance and genus richness with artificial data (representing these missing genera) augmented onto the real data. We augment the data with the real species richness of each missing genus and an average per-species abundance of 1. The analysis of these augmented data yields the same answer: the hump-shaped relationship between average per-species abundance and genus richness is strongly supported by a likelihood ratio test

```
# what if we had a bunch of rare, low richness genera
gen_what_if <- rbind(select(gen, c(genus, nspp, log_abund)),
  data.frame(genus = paste0("fake_", 1:n_checklist_missing),
    # actual abundances of missing genera
    nspp = 1, #gen_rich$nspp[gen_rich$status == "end"],
    # log(1) = 0
```

```

log_abund = 0))

# quadratic model
what_if_mod <- glm_bgeom(nspp ~ log_abund + I(log_abund^2), data = gen_what_if)

# straight line model
what_if_mod0 <- glm_bgeom(nspp ~ log_abund, data = gen_what_if,
                          a_init = c(1, 1))

lrt_bgeom(what_if_mod0, what_if_mod)

```

Likelihood ratio test

Model 0: nspp ~ log_abund

Model 1: nspp ~ log_abund + I(log_abund^2)

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
Model 0	3	-707.37			
Model 1	4	-675.57	1	63.599	1.525e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

S5.5 Assessing the choice of average per-species abundance

We should evaluate if the choice of average per-species abundance produces any unintended artifacts in our analysis of how average per-species abundance relates to within genus richness. The primary concern would be if genera themselves somehow had a carrying capacity, then the more species contained in a genus, the less abundant, on average, each species would have to be. Two observations from the data dispel this concern: first, the vast majority of genera are represented by only a single species at a sampling location Supplementary Figure S7, making the partitioning of a hypothetical genus-level carrying capacity moot at all but the few sites where genera are represented by multiple species.

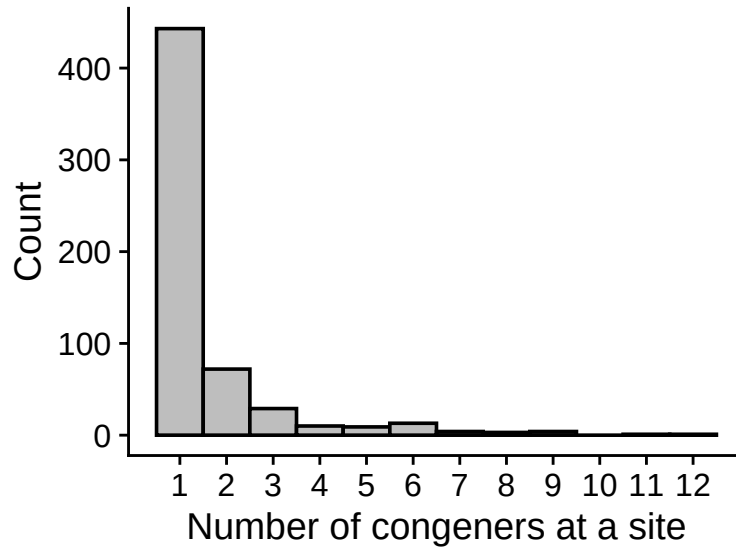
```

# computing number of species per genus and total genus abundance per site
gen_by_site <- summarize(arth,
  rich = n_distinct(species_code),
  tot_abund = n(),
  .by = c(site, genus))

ggplot(gen_by_site, aes(rich)) +
  geom_histogram(binwidth = 1, color = "black", fill = "gray") +

```

```
scale_x_continuous(breaks = 1:12) +
xlab("Number of congeners at a site") +
ylab("Count") +
theme_cowplot()
```



Supplementary Figure S7: Distribution of numbers of species per genus observed in survey data (Gruner 2007).

Next, if a hypothetical genus carrying capacity were partitioned across multiple species, then at sites with multiple congeners there should be a flat relationship between number of congeners and total genus abundance per site. However, Supplementary Figure S8 shows there is a positive relationship between number of congeners and total genus abundance, suggesting that the abundance of any one species in a genus does not trade-off with the abundances of other congeners, but rather they all add to a larger total genus abundance. The positive relationship is significant by a likelihood ratio test applied to a Poisson GLM of abundance predicted by log number of congeners (note: excluded are any data points in which only a single species in a genus is present at a site).

```
gen_site_mod <- glm(tot_abund ~ log(rich),
                    data = gen_by_site[gen_by_site$rich > 1, ],
                    family = "poisson")

anova(gen_site_mod)
```

Analysis of Deviance Table

Model: poisson, link: log

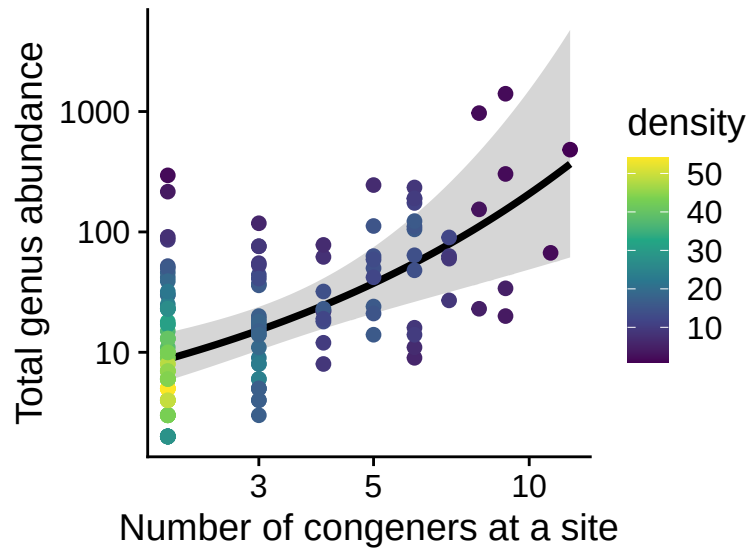
Response: tot_abund

Terms added sequentially (first to last)

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			145	20347	
log(rich)	1	9252.3	144	11095	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
ggplot(gen_by_site[gen_by_site$rich > 1, ],
       aes(rich, tot_abund)) +
  geom_smooth(color = "black",
             method = "glm",
             method.args = list(family = "poisson")) +
  geom_pointdensity() +
  scale_x_log10() +
  scale_y_log10() +
  scale_color_viridis_c() +
  xlab("Number of congeners at a site") +
  ylab("Total genus abundance") +
  theme_cowplot()
```



Supplementary Figure S8: Relationship between number of congeners at a site and total genus abundance at that site. Trend line (black line) and 95% confidence envelope (gray shaded region) are from a Poisson GLM. Data points are colored by local point density.

This analysis of number of congeners and total genus abundance at the scale of individual sampling locations is distinct from the analysis of average per-species abundance and total genus richness at the scale of the archipelago, the latter being the one that found a hump-shaped relationship between abundance and richness. The sampling location-scale analysis is intended to evaluate a possible genus-level carrying capacity at the scale of an individual site. The sampling location-scale analysis calculates abundance on a site-by-site basis (not averaged across all sites as in the archipelago-wide analysis) and uses the observed number of congeners at a site (not the total richness of a genus from across the archipelago).

References

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- Gruner, D. S. 2007. Geological age, ecosystem development, and local resource constraints on arthropod community structure in the hawaiian islands. - *Biological Journal of the Linnean Society* 90: 551–570.
- McGillcuddy, M., Warton, D. I., Popovic, G. and Bolker, B. M. 2025. [Parsimoniously fitting large multivariate random effects in glmmTMB](#). - *Journal of Statistical Software* 112: 1–19.

Nishida, G. M. 2002. Hawaiian terrestrial arthropod checklist.